



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences
Second Year - Semester I Examination - August/September 2024

MAT 2305 - REAL ANALYSIS II

Time: Three (03) hours.

Answer ALL (06) questions.

1. a) Find: $\sum_{n=1}^{\infty} \frac{3^n - 5^{n+2}}{6^{n-1}}$.

(25 marks)

b) Use the comparison test for the convergence of infinite series to show that the following series are convergent:

i. $\sum_{n=1}^{\infty} \frac{1}{n^2}$

ii. $\sum_{n=1}^{\infty} \sqrt{n^{16} + 1} - \sqrt{n^{16} - 1}$

(50marks)

c) Determine the convergence of the series: $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^{1.5}}$ using the integral test.

(25 marks)

2. a) State the limit form of the ratio test for the convergence of infinite series.

Use the ratio test to determine the convergence or divergence of each of the following series:

- i. $\sum_{n=1}^{\infty} \frac{5^n}{n!}$
- ii. $\sum_{n=1}^{\infty} \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{n!}$
- iii. $\sum_{n=1}^{\infty} \frac{n!}{(3n)!}$

(65 marks)

- b) Test the convergence of the following series using the root test:

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{2n+7}{8n-2} \right)^{2n}$$

(35 marks)

3. a) Consider the power series: $\sum_{n=0}^{\infty} a_n(x-c)^n$ with non-zero coefficients a_n .

Show that its radius of convergence R is

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|,$$

provided the limit exists.

Determine the radius of convergence and the interval of convergence for the power series:

$$\sum_{n=0}^{\infty} \frac{n^3 x^n}{n!}.$$

(50 marks)

- b) Write down the power series expansion of $\frac{1}{x+1}$ for $|x| < 1$.

Hence, show that

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \quad -1 < x \leq 1.$$

Deduce that

$$\frac{1}{2} (\ln(1+x))^2 = \frac{x^2}{2} - \frac{x^3}{3} \left(1 + \frac{1}{2} \right) + \frac{x^4}{4} \left(1 + \frac{1}{2} + \frac{1}{3} \right) - \cdots, \quad -1 < x \leq 1.$$

(50 marks)

4. a) Evaluate the limit : $\lim_{(x,y) \rightarrow (0,0)} \frac{y \sin(x^2 + y^2)}{x^2 + y^2}$.
(20 marks)
- b) Suppose $z = e^y \cos x$, $x = s^2 t$, and $y = st^3$. Find $\frac{\partial z}{\partial t}$.
(20 marks)
- c) Determine all critical points of the function $f(x, y) = x^3 + y^3 - 3xy$.
Discuss the nature of each critical point.
(60 marks)
5. Use the Lagrange multiplier method to solve the following problems:
- a) Find the minimum and maximum values of the function: $f = xyz$ subject to the constraint:
 $x^2 + y^2 + z^2 = 1$.
(60 marks)
- b) Find the largest product of the numbers a and b when $a^2 + b^2 = 1$.
(40 marks)
6. a) Evaluate the integral: $\iint_R (2x + 3y) dA$, where R is the rectangular region in $\mathbb{R}^2$ defined by
 $0 \leq x \leq 2$ and $1 \leq y \leq 3$.
(30 marks)
- b) Evaluate the integral: $\iint_R e^{x^2 + y^2} dA$, where $R = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x^2 + y^2 \leq 2\}$.
(35 marks)
- c) Evaluate the integral: $\iiint_D (x^2 + y^2) dV$, where D is the cylinder in $\mathbb{R}^3$ defined by
 $0 \leq x^2 + y^2 \leq 4$ and $0 \leq z \leq 3$.
(35 marks)

—————END—————